

LIGHT POSTERIOR EXCHANGE FOR COMMUNICATION-EFFICIENT DISTRIBUTED LMB FUSION

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ABSTRACT

Distributed labeled multi-Bernoulli (LMB) fusion often exchanges full Gaussian-mixture posteriors even when the receiver-side fusion step ultimately consumes label-wise moment-matched Bernoulli statistics. This paper studies a simpler communication lever: keep the effective Kullback-Leibler average (KLA) fusion graph intact, but replace full posterior messages with light posteriors containing per-label existence probabilities and single-Gaussian moments. On a 4+4 multisensor formation benchmark, a periodic light-posterior exchange matches the full-posterior baseline in tracking and consensus metrics over 50 held-out trials while reducing estimated communication by 58.6%. The result suggests that, for single-round GA/KLA-LMB fusion, payload compression can be safer than event-triggered graph sparsification.

Index Terms— Distributed multi-object tracking, labeled random finite sets, LMB filter, KLA fusion, communication-efficient sensing

1. INTRODUCTION

Distributed multi-object tracking lets a sensor network combine complementary fields of view without routing all measurements to a central processor. Labeled random finite-set filters, and in particular the labeled multi-Bernoulli (LMB) family, are attractive in this setting because they maintain object existence, kinematic uncertainty, and track identity in a single posterior representation [1, 2, 3]. The practical difficulty is communication. A multi-sensor LMB posterior can contain many labels and multiple Gaussian components per label, so repeatedly broadcasting full Gaussian-mixture posteriors is expensive for bandwidth-limited platforms.

A common response is to communicate less often or with fewer neighbors. This is appealing, but it can attack the wrong bottleneck. Conservative fusion rules such as covariance intersection, generalized covariance intersection, and Kullback-Leibler average (KLA) fusion are designed to combine partially dependent estimates without requiring unknown cross-correlations [4, 5]. In a distributed tracker, however, the quality of the fused posterior depends not only on whether the nominal graph is

connected, but also on which posterior messages actually enter the fusion step at each time. If event triggers or topology adaptation remove too many effective message paths, consensus and label agreement can degrade even while byte counts improve.

This paper takes a different position: preserve the effective fusion graph first, then reduce the payload carried by each message. The key observation is implementation-driven. In many practical GA/KLA-LMB fusion pipelines, incoming LMB components are aligned by label and reduced to label-wise Bernoulli moments before fusion. If the downstream fusion rule consumes these moments, transmitting every Gaussian-mixture component may be unnecessary in regimes where a single Gaussian moment is an adequate summary. This suggests a light posterior message containing the existence probability, mean, covariance, and minimal label metadata for each active Bernoulli component.

We study this idea in a distributed 4+4 multisensor formation benchmark. The baseline periodically exchanges full LMB posteriors over the static communication graph. The proposed light-posterior exchange keeps the same graph and synchronization schedule, but replaces each full Gaussian-mixture label density with its moment-matched single-Gaussian approximation. This design deliberately avoids graph sparsification: every baseline information path remains available to the KLA update, while communication is reduced by changing the representation of each label.

The empirical result is simple but instructive. Over 50 held-out trials, periodic light-posterior exchange reduces estimated communication by 58.6% relative to periodic full-posterior exchange, while matching the baseline local E-OSPA, consensus OSPA, position-disagreement, cardinality-dispersion, and effective-weight graph connectivity metrics. In contrast, dynamic-topology variants that alter the communication graph are less stable in the same benchmark. These results support a narrow claim: for single-round distributed GA/KLA-LMB fusion with label-wise moment matching, full posterior exchange can be overkill, and graph-preserving payload compression is a safer first lever than sparse event-triggered communication.

The contributions are:

- a graph-preserving light posterior exchange protocol for distributed LMB fusion, using label-wise moment-matched Bernoulli messages instead of full Gaussian-mixture posterior messages;
- an effective-fusion-graph perspective that separates payload compression from message-path sparsification; and
- a held-out 50-trial validation showing full-posterior-equivalent tracking and consensus with about 58% lower estimated communication in a 4+4 multisensor benchmark.

2. RELATED WORK

Labeled random finite-set tracking. Random finite-set filters provide a principled Bayesian framework for multi-object tracking with uncertain cardinality and measurement association. The probability hypothesis density filter introduced a tractable first-moment recursion for multi-object tracking [6], while labeled RFS models attach persistent identities to objects [1]. The LMB filter gives a computationally attractive labeled approximation with independent Bernoulli components [2], and multi-sensor GLMB/LMB variants extend labeled RFS tracking to distributed and cooperative sensing settings [3]. Our work does not change the LMB recursion itself; it studies how much posterior detail must be communicated before a distributed KLA-style fusion step.

Conservative distributed fusion. When local posteriors are statistically dependent and cross-correlations are unknown, conservative fusion rules are used to avoid overconfident estimates. Covariance intersection is a classical solution for Gaussian estimates [4], and KLA or generalized covariance intersection extends this log-opinion-pool principle to probability densities [5, 7]. Several RFS fusion methods apply KLA/GCI ideas to multi-Bernoulli or LMB densities, including exponential-mixture PHD fusion [8], multi-Bernoulli GCI fusion [9], and robust distributed labeled RFS fusion [10, 11]. These methods primarily define how received posteriors should be fused. We focus on the complementary question of what posterior representation should be transmitted when the fusion implementation already performs label-wise moment reduction.

Communication-aware RFS fusion. Communication constraints have been addressed through selective transmission, adaptive scheduling, consensus design, and alternative density representations. Communication-constrained decentralized fusion has been studied for labeled RFS trackers [12], consensus LMB filters reduce reliance on a central node [13], and recent adaptive fusion work adjusts weights or reliability scores to account for heterogeneous sensing and communication conditions

[14, 15]. Other work studies resilient fusion under unreliable communication or approximate average-density fusion [16]. These lines often reduce communication by changing when nodes communicate, which neighbors they use, or how much trust a received posterior receives. Our experiments show why those graph-level choices can be fragile for single-round KLA fusion: preserving the effective message graph may matter more than reducing the number of message paths.

Payload compression versus graph sparsification. The closest conceptual distinction in this paper is between transmitting fewer posterior paths and transmitting lighter posterior summaries along the same paths. Event-triggered or topology-adaptive communication can save bytes, but it also changes the effective fusion graph seen by the KLA update. A light posterior message instead leaves the graph unchanged and compresses each label density to the statistics consumed by the receiver-side fusion path. This makes the communication saving easier to attribute: any tracking or consensus change is due to posterior summarization, not to missing neighbors. The held-out results indicate that, in the tested GA/KLA-LMB setting, this payload-first design is sufficient to recover full-posterior behavior with substantially lower communication.

3. LIGHT POSTERIOR EXCHANGE

This section will define the full-posterior baseline, the label-wise moment-matched light posterior message, and the graph-preserving exchange protocol.

4. EXPERIMENTS

This section will report the 4+4 multisensor formation benchmark, communication accounting, effective-fusion-graph diagnostics, and held-out 50-trial results.

5. CONCLUSION

This section will summarize when light posterior exchange is appropriate and when richer Gaussian-mixture messages or adaptive communication may still be needed.

6. REFERENCES

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